

E-Commerce Sales & Customer Analytics

Power BI Dashboard Specification

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Version	1.0
Dashboard Pages	3
Audience	Sales, Marketing, Operations, Finance, Product

Portfolio note: This is a simulated dashboard specification; the actual PBIX can be built using the supplied dataset and DAX.

1. Dashboard Objective

Create a governed, interactive management dashboard that converts sales and customer data into actionable performance insights.

2. Page 1 — Executive Sales Dashboard

Component	Design / Requirement
KPI Cards	Total Revenue, Completed Orders, AOV, Return Rate, Unique Customers.
Revenue Trend	Line chart by Year Month.
Regional Performance	Bar chart sorted by revenue descending.
Category Performance	Column chart comparing category revenue.
Revenue Share	Donut chart showing category contribution.
Top Products	Top 10 products by completed-order revenue.
Slicers	Date, Region, Category, Product, Order Status.

3. Page 2 — Customer Analysis

Visual	Business Purpose
New vs Repeat	Understand customer retention mix.
Top Customers	Identify high-value customers.
Revenue per Customer	Measure customer value.
Orders per Customer	Understand engagement frequency.
Customer trend/distribution	Identify concentration and opportunity.

4. Page 3 — Returns & Operations

Visual	Business Purpose
Return Rate by Category	Identify categories requiring investigation.
Returns by Region	Identify geographic patterns.
Return Trend	Monitor change over time.
Category Comparison	Prioritize operational root-cause analysis.

5. Data Model

Use `ecommerce\_sales\_data` as the fact table and a dedicated Calendar dimension for time intelligence. The relationship should be one-to-many from Calendar[Date] to ecommerce_sales_data[order_date].

6. Core DAX Measures

Measure	Purpose
Total Revenue	Completed-order revenue.
Completed Orders	Count of completed orders.
Average Order Value	Revenue divided by completed orders.
Return Rate	Returned orders divided by qualifying orders.
Unique Customers	Distinct completed-order customers.
Revenue per Customer	Revenue divided by unique customers.
MoM Revenue %	Month-over-month revenue change.

7. Interaction Requirements

Slicers should cross-filter applicable visuals. Category should support drill-down to product. Tooltips should provide supporting KPIs. Date selections should update all relevant pages.

8. UAT Acceptance Criteria

- Dashboard KPIs reconcile to approved SQL/Excel calculations.
- Date, region and category filters work consistently.
- Drill-down preserves filter context.
- Refresh timestamp is visible.
- No visual produces contradictory totals.

9. BA Interview Explanation

Describe Power BI as the presentation and decision-support layer. As BA, your responsibility is to define KPI logic, dashboard requirements, dimensions, interactions, business questions and acceptance criteria; BI developers then implement the model and visuals.